Khashayar Pourtaheri

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Education

University of Washington — Expected 2027
Ph.D. in Economics | Advisor: Alan Griffith — Seattle, WA, USA

Smith School of Business, Queen's University — 2021
M.Sc. in Management, Business Economics — Kingston, ON, Canada

Sharif University of Technology — 2020
M.Sc. in Economics

Sharif University of Technology — 2017
B.Sc. in Mechanical Engineering

Research

Fields: Applied Microeconomics, Health and Labor Economics, Causal Inference, Behavioral Economics and AI

Working Papers

The Social Cost of Drug Decriminalization: Overdoses, Crime, and Recriminalization in Oregon and Washington [link]
Job Market Paper | SSRN working paper — October 2026

What does society gain and lose when states reduce criminal penalties for drug possession? Using FBI arrest and crime records and CDC death records, I compare Oregon's and Washington's 2021 reforms with weighted comparison states that tracked them before 2021. Drug-user arrests fall 68 percent with no shift to other charges, while overdose deaths rise 34 percent and property crime 17 percent. Deaths rise in every comparison group, including groups matched on COVID, fentanyl's spread and state politics. Restoring penalties in Washington brings arrests back but not deaths down. Health and crime costs outweigh the gains from less enforcement: a net social cost of $26.6 billion by mid-2023, mostly from 1,971 excess deaths. The results show that reducing penalties for drug possession can cut arrests sharply and still leave society worse off.

Behavioral History Outperforms Descriptions of the Person for LLM Synthetic Personas [link]
with Ahmad Zareei | arXiv preprint — October 2026

Large language models (LLMs) are increasingly used as synthetic personas representing survey respondents. Their validity as substitutes for particular respondents depends on whether they reproduce individuals' decisions. We examine what information helps synthetic respondents predict each individual's later choices, using five conditions that add progressively richer information: no personal information, demographics, personality traits, cognitive scores, and finally the respondent's earlier survey choices as behavioral history. We use a two-wave panel of 845 US adults who completed measures of 14 behavioral biases (spanning risk, time preferences, overconfidence, and reasoning), so each respondent's earlier answers provide a human test–retest benchmark; in the behavioral-history condition, all items that score the target bias are withheld. At the population level, the average number of biases per respondent in every condition is close to the human average (7.1–8.1 biases, against 7.1 for humans). This aggregate similarity masks differences in variance: persona descriptions recover only 53–67% of human between-person variation, whereas adding behavioral history restores it to approximately the human level. At the individual level, description-based personas achieve only 7–12% of the informedness observed in human test–retest responses, while adding behavioral history raises this to 28%. The condition including behavioral history has the highest estimated informedness in all 17 demographic groups, whereas description-based conditions provide little or no information for some groups. Synthetic responses also exhibit stronger education- and income-related differences than human responses. For LLM synthetic personas, a respondent's past answers add more to individual-level prediction than a description of who they are.

Work in Progress

- **The Effects of Common Ownership on Oligopoly**
  Contract-theoretic model of the conditions under which common-ownership links can sustain collusive product-market outcomes.
- **The Price Effect of the Salon Mechanism in a Freight Transportation Market**
  Reduced-form analysis of 40 million bills of lading estimating the price effects of government intervention and information disclosure in the freight market.

Work Experience

Research Economist Sep 2026 – Present
Comparative Health Outcomes, Policy, and Economics (CHOICE) Institute
Project: Adolescent Mental Health Study | Supervisor: Noémi Kreif

- Audited the causal framework, R estimation pipeline, and outcome coding of a survey-weighted, cross-fitted TMLE study of adolescent depression and adult labor outcomes (restricted-use Add Health), documenting coding errors and proposing sensitivity analyses (E-values, MNAR).
- Reviewed 46 studies and built a DAG to identify the factors that drive both mental health and employment, then designed a selection strategy that combines the DAG with data-driven methods to keep every key confounder among the more than 6,000 variables in the data.
- Translated treatment effects into labor-supply elasticities for macroeconomic and ROI models, correcting for school enrollment and avoiding double counting across hours, participation, and earnings.

Data Scientist Researcher — Institute for Health Metrics and Evaluation (IHME) Summer 2025

- Produced subnational estimates of HIV prevalence, antiretroviral-therapy coverage, and viral-load suppression for Malawi and Zimbabwe by integrating PHIA, DHS, and AIS survey data in reproducible pipelines.
- Conducted statistical, machine-learning, and forecasting analyses; documented legacy code and mapped the end-to-end workflow.

Research Assistant — Smith School of Business, Queen's University May 2021 – Aug 2021

- Integrated and cleaned data from multiple sources, validated data quality, and communicated statistical findings through reproducible analysis in R and Stata.

Teaching

Instructor — Intro to Microeconomics, University of Washington Winter 2024

- Designed and delivered lectures on supply and demand, market equilibrium, consumer theory, and firm behavior; created assessments and held office hours.

Teaching Assistant — University of Washington Sep 2022 – Present

- Courses: econometrics and data science; probability and statistics; introductory micro- and macroeconomics.
- Taught R and Python labs on data cleaning, visualization, reproducible workflows, supervised learning, cross-validation, regularization, tree-based models, and neural networks.

Teaching Assistant — Managerial Economics (MBA), Queen's University Jan 2021 – May 2021

- Led quiz and review sessions for MBA students, translating economic concepts for working professionals.

Skills

Econometrics: Event-study DiD, synthetic DiD and synthetic control, DML, CV-TMLE, AIPW, and Super Learner, IV, RDD, survey-weighted inference, mediation and sensitivity analysis (E-values, MNAR)
Programming: R (tmle, SuperLearner, survey), Stata, Python, SQL, Git, MATLAB, LaTeX/Beamer
Data: Add Health (restricted-use), NLSY97, FBI NIBRS, CDC WONDER, PHIA, DHS, AIS
Other: LLM experiment design (Claude API), machine learning (scikit-learn, PyTorch), Tableau

References

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